EE3032 Introduction to VLSI Design—Midterm

1. (40%) Answer the following problems.
   (a) Draw a CMOS transistor-level schematic realizing the Boolean function $f = \overline{a(b+c)}$. (4%)
   (b) List the process for transferring the mask pattern to the surface of a silicon region. (4%)
   (c) List three approaches for increasing the performance of a CMOS circuit. (6%)
   (d) List two latchup prevention approaches. (4%)
   (e) List three principles for reducing the switching power of a CMOS circuit.(6%)
   (f) Explain the purpose of design rules in VLSI design. (4%)
   (g) Explain what the body effect is. (4%)
   (h) Explain why adding a guard ring can reduce the latchup effect. (4%)
   (i) Explain the energy consumption of an inverter when the inverter performs a switching. (4%)
2. (10%) (a) Design an 8-to-1 multiplexer using pass transistor logic. (5%) (b) Design a two-input exclusive-or (XOR) gate using CMOS transmission gates. (5%)
3. (20%) Complete the following problems.
   (a) Figure 1(a) shows the XOR and OR gate. Assume that the P_A and P_B are the mutually independent signal probabilities of the inputs A and B to be at "1". Express the $P_{0\to1}$ of X and Y. (5%)
   (b) Figure 1(b) shows a circuit with one OR and two AND gates. Assume the 1's probabilities of inputs all are 0.2. Calculate the $P_{0\to1}$ of A, B, and C. (10%)
   (c) Figure 1(c) shows the pass transistor logics with NMOS and PMOS. Level the source and drain of the circuits, and calculate the V_{out} of the circuits. (5%)

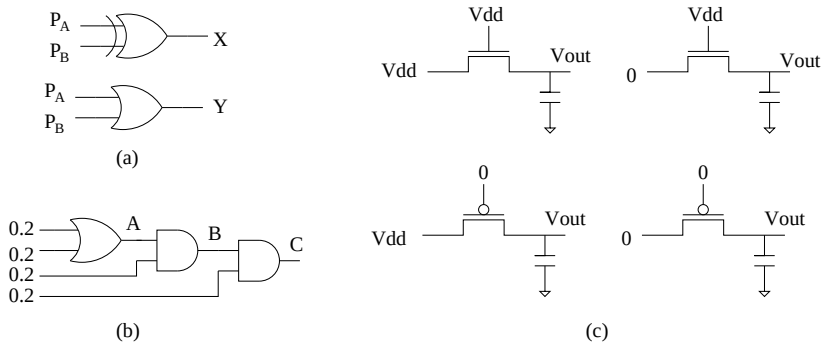


Figure 1: (a) primitive gates (b) cascade design (c) pass transistors.

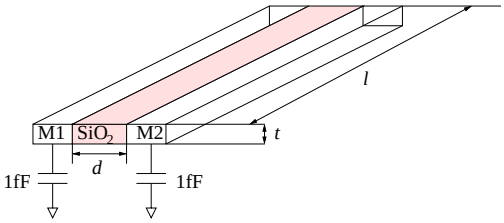


Figure 2: Two parallel wires.

4. (10%) There are two parallel wires, M1 and M2, as shown in Figure 2. Assume that $\varepsilon_{SiO_2} = 5 \times 10^{-14} F/cm$, $d = 1\mu m$, $t = 1\mu m$, and $l = 100\mu m$. If the M2 is floating and M1 has a 0V to 1.8V transition, how much crosstalk noise will be coupled on the M2?

5. (10%) Figure 3 depicts two designs of inverter chains. Assume that the equivalent resistance of a minimum size W/L NMOS is R and $\mu_n = 2\mu_p$. The gate capacitance of a W/L transistor is C_g. Also, the parasitic capacitances of drain/source and wires are neglected. Which inverter chain consumes lower short-circuit power. Explain your answer.

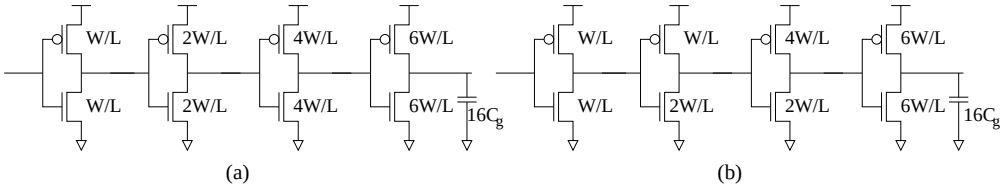
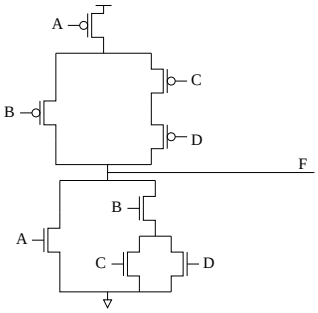


Figure 3: Inverter chains.

6. (10%) Suppose that the signal delay on an interconnect of length $50\mu m$ is known to be $0.13ps$. If the line is increased to $100\mu m$, then what is the delay of the line? (4%) If you can insert an inverter in the $100\mu m$ interconnect and the inverter delay is $0.2ps$, where should the inverter be inserted such that the wire delay is minimized, and how much is the delay of the interconnect with buffer insertion? (6%)

7. (10%) Size the transistors in the circuit shown below that implements a function $F = \overline{A + B(C + D)}$ so that its worst case current driving capability is equivalent to that of an inverter with $W_p/L_p = 10\lambda/2\lambda = 5$ and $W_n/L_n = 4\lambda/2\lambda = 2$. Assume that the channel length of all transistors is fixed at 2λ.


A
C
B
D
F
B
A
C
D